

Time Translations and Hamiltonian Dynamics

A guided derivation of why the Hamiltonian generates time evolution

Student release

2026-05-08

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! How to use this packet

This is not an introductory reading. It is a guided derivation for a student who already knows the basic quantum-mechanical statement that momentum generates spatial translations and is comfortable doing symbolic calculations with Taylor series, partial derivatives, kinetic energy, potential energy, and Newton's second law.

Do not read this passively. The problems are the packet. Keep a scratch notebook open and work through the derivations as you read.

The goal is to understand, from the classical side, why the Hamiltonian is the object that generates time evolution.

In quantum mechanics one often says:

1. momentum generates translations in space,
2. the Hamiltonian generates translations in time.

The first statement is relatively direct. The second statement leads through the Hamiltonian formulation of classical mechanics and then points toward quantization.

The route through the packet is:

1. review spatial translations as the model example,
2. infer the dimensions of the generator of time translations,
3. recover canonical momentum from kinetic energy,
4. build the Hamiltonian by a Legendre transform,
5. derive Hamilton's equations,
6. introduce the Poisson bracket,
7. recognize the Poisson bracket as a Lie bracket,
8. handle explicit time-dependence,
9. connect the classical story to the commutator in quantum mechanics.

Useful external references:

- [Hamiltonian mechanics](#), for the phase-space formulation of classical dynamics.
- [Legendre transformation](#), for the mathematical operation that replaces velocity by momentum.
- [Poisson bracket](#), for the bracket structure on phase-space functions.
- [Lie algebra](#), for the abstract algebraic structure behind brackets satisfying the Jacobi identity.
- [Canonical quantization](#), for the later replacement of Poisson brackets by commutators.

Throughout the main derivation, we work mostly in one spatial dimension to keep the notation light. The same structure extends to many degrees of freedom by replacing (x, p) with (x^i, p_i) .

1 Spatial translations as the model example

In class we learned that the linear momentum p is the generator of translations in space. Concretely, from p we can build an operator $U(a)$ on functions of position such that

$$U(a)f(x) = f(x + a).$$

Translations should preserve the Hilbert-space norm, so $U(a)$ should be unitary. With the usual position-space representation

$$p = -i\hbar \frac{d}{dx}$$

we get

$$\frac{i}{\hbar}ap = a \frac{d}{dx}.$$

Thus the translation operator may be written as

$$U(a) = \exp\left(\frac{i}{\hbar}ap\right) = \exp\left(a \frac{d}{dx}\right).$$

i Problem 0: The exponential translation operator

Show explicitly that

$$\exp\left(a \frac{d}{dx}\right) f(x) = f(x + a)$$

by expanding the exponential and reinterpreting the result as a Taylor series.

This calculation is the model for the rest of the packet. We now ask for the analogous generator of translations in time.

2 Guessing the generator of time translations

We want an operator $U(c)$ that maps a function of time to its translated version:

$$U(c)f(t) = f(t + c).$$

By the same Taylor-series argument,

$$U(c) = \exp\left(c \frac{d}{dt}\right).$$

Write this in the quantum-mechanical form

$$U(c) = \exp\left(-\frac{i}{\hbar}ch\right),$$

where h is a Hermitian operator. The minus sign is conventional and matches the usual Schrödinger time-evolution convention.

Comparing the two expressions gives

$$c \frac{d}{dt} = -\frac{i}{\hbar}ch.$$

Thus h is the object that should generate translations in time.

i Problem 1: Dimensions of the time-translation generator

Find the physical dimensions of the generator h . What familiar real physical quantity has these dimensions?

The dimensional argument strongly suggests that the generator of time translations is the Hamiltonian. To see why, we need to revisit the relationship between kinetic energy, momentum, and Newton's second law.

3 Momentum from kinetic energy

Newton's second law is

$$F = \frac{dp}{dt}.$$

In Hamiltonian mechanics, momentum is not merely mv . More generally, momentum is defined as the derivative of kinetic energy with respect to velocity.

i Problem 2: Momentum as a derivative of kinetic energy

For a free particle of mass m moving in Euclidean space, the kinetic energy is

$$K = \frac{1}{2}m\mathbf{v} \cdot \mathbf{v}.$$

Show that

$$p_i = \frac{\partial K}{\partial v^i}.$$

Equivalently,

$$\mathbf{p} = \frac{\partial K}{\partial \mathbf{v}}.$$

In one dimension, this becomes

$$p = \frac{\partial K}{\partial v}.$$

This definition is the entrance to Hamiltonian mechanics.

4 The Legendre transform and the Hamiltonian

For a conservative force, there is a potential energy $U(x)$ such that

$$F = -\frac{\partial U}{\partial x}.$$

The usual total energy is

$$E = K + U.$$

However, Hamiltonian mechanics wants to describe dynamics using position and momentum, not position and velocity. We therefore want an energy function of x and p , not x and v .

The correct transformation from velocity to momentum is the **Legendre transform**. In one dimension, define

$$H(x, p) = pv - K(x, v) + U(x),$$

where

$$p = \frac{\partial K}{\partial v}.$$

The velocity v is then to be regarded as the velocity corresponding to the chosen value of p .

i Problem 3: The Legendre transform removes velocity dependence

Treat p and v as independent variables during the partial-derivative calculation, while imposing

$$p = \frac{\partial K}{\partial v}.$$

1. Show that

$$H(x, p) = pv - K(x, v) + U(x)$$

is independent of v .

2. Suppose

$$K(x, v) = f(x)v^\alpha$$

for some real number α . Show that

$$pv - K = (\alpha - 1)K.$$

3. Conclude that if K is quadratic in v , then

$$H = K + U.$$

This is why the Hamiltonian is often equal to the total energy. But the definition

$$H = pv - K + U$$

is more fundamental than the slogan $H = K + U$, because the Hamiltonian is supposed to be a function on phase space.

5 Phase space and Hamilton's equations

The configuration space is the space of configurations. When we are talking about a point particle, it is the space of positions. In one dimension it is coordinatized by x . In n dimensions it is coordinatized by

$$x^1, \dots, x^n.$$

The phase space is the space of positions and momenta. In one dimension it has coordinates

$$(x, p).$$

In n dimensions it has coordinates

$$(x^1, \dots, x^n, p_1, \dots, p_n).$$

Thus if configuration space is n -dimensional, phase space is $2n$ -dimensional.

Hamiltonian mechanics says that physical motion is a curve in phase space.

For a particle of mass m in one dimension,

$$K = \frac{1}{2}mv^2, \quad p = mv.$$

Solving for v ,

$$v = \frac{p}{m}.$$

Thus

$$K = \frac{p^2}{2m}.$$

The Hamiltonian is therefore

$$H(x, p) = \frac{p^2}{2m} + U(x).$$

i Problem 4: The first Hamilton equation

Use

$$H(x, p) = \frac{p^2}{2m} + U(x)$$

to show that

$$\frac{dx}{dt} = \frac{\partial H}{\partial p}.$$

i Problem 5: The second Hamilton equation

For the same particle, use Newton's second law and

$$F = -\frac{\partial U}{\partial x}$$

to show that

$$\frac{dp}{dt} = -\frac{\partial H}{\partial x}.$$

Together,

$$\boxed{\frac{dx}{dt} = \frac{\partial H}{\partial p}, \quad \frac{dp}{dt} = -\frac{\partial H}{\partial x}.$$

These are **Hamilton's canonical equations**. They say that the Hamiltonian function determines motion through phase space.

6 Phase-space functions and the Poisson bracket

Let $f(x, p)$ be any differentiable function on phase space. For example, f might be x , p , H , xp , $e^{i\frac{xp}{\hbar}}$ or some more complicated quantity.

i Problem 6: Time evolution of a phase-space function

Use the chain rule and Hamilton's equations to show that

$$\frac{df}{dt} = \frac{\partial f}{\partial x} \frac{\partial H}{\partial p} - \frac{\partial f}{\partial p} \frac{\partial H}{\partial x}.$$

This expression is important enough to deserve its own notation. For two phase-space functions $f(x, p)$ and $g(x, p)$, define the **Poisson bracket**

$$\{f, g\} = \frac{\partial f}{\partial x} \frac{\partial g}{\partial p} - \frac{\partial f}{\partial p} \frac{\partial g}{\partial x}.$$

i Problem 7: Hamiltonian flow as a Poisson bracket

Use the definition of the Poisson bracket to rewrite the result of Problem 6 as

$$\frac{df}{dt} = \{f, H\}.$$

Then compute the basic brackets

$$\{x, p\}, \quad \{x, x\}, \quad \{p, p\}.$$

This is the classical algebraic skeleton that later forms the foundation of quantum mechanics.

7 The Poisson bracket as a Lie bracket

The Poisson bracket was not added by hand. Once we describe dynamics on phase space using $H(x, p)$, the bracket appears naturally.

A bracket operation is called a **Lie bracket** if it is bilinear, antisymmetric, and satisfies the Jacobi identity. We will now define these things and check that the Poisson bracket on space $\mathcal{H}$ of smooth functions on the phase space is a Lie bracket.

i Problem 8: Bilinearity and antisymmetry

For phase-space functions f , g , and h , show that the Poisson bracket satisfies:

1. Bilinearity:

$$\{af + bg, h\} = a\{f, h\} + b\{g, h\}$$

for constants $a, b \in \mathbb{R}$, and similarly in the second slot.

2. Antisymmetry:

$$\{f, g\} = -\{g, f\}.$$

The third Lie-bracket identity is the Jacobi identity.

i Problem 9: The Jacobi identity for the Poisson bracket

Show, or at least check in a convincing way, that the Poisson bracket satisfies the Jacobi identity:

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0.$$

This is an algebraic identity, not an equation of motion.

A structure that allows you to multiply its elements together is called an **algebra**. A **Lie algebra** is a vector space that forms an algebra under a bilinear operation that is also a Lie bracket. The phase-space functions form a vector space because you can add functions, multiply functions by numbers, it has a zero function, *et cetera*. Since all this is coming from the Poisson bracket, we give it the special name **Poisson algebra**: an ordinary commutative algebra of functions equipped with the Lie-bracket structure supplied by the Poisson bracket.

The Hamiltonian is now not merely an energy function. It is the generator of motion through this algebra:

$$\boxed{\frac{df}{dt} = \{f, H\}.$$

This is the classical version of the statement that time translations are generated by the Hamiltonian.

8 Explicit time dependence and conservation of energy

The equation

$$\frac{df}{dt} = \{f, H\}$$

assumes that f depends on time only through $x(t)$ and $p(t)$. But a phase-space function may also depend explicitly on time. Then

$$f = f(x, p, t).$$

i Problem 10: Explicit time-dependence

For a function $f(x, p, t)$ on extended phase space, show that

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \{f, H\}.$$

Then use this result to show that the Hamiltonian is conserved when it has no explicit time-dependence.

This completes the classical derivation. The Hamiltonian generates time translations because it generates the time evolution of every phase-space function through the Poisson bracket.

9 Epilogue: toward Hamiltonian quantization

In classical Hamiltonian mechanics, the phase-space functions form an algebra equipped with the Poisson bracket. The Poisson bracket is bilinear, antisymmetric, and satisfies the Jacobi identity, so it behaves like a Lie bracket.

Quantum mechanics replaces classical phase-space functions by operators. Operators can be added and multiplied, but their multiplication is generally not commutative. This gives us a new natural bracket: the commutator.

i Problem 11: The commutator Jacobi identity

Suppose $\mathcal{O}$ is an algebra of operators with an associative multiplication. This means that for any $A, B, C \in \mathcal{O}$,

$$A(BC) = (AB)C.$$

Define the **commutator** by

$$[A, B] = AB - BA.$$

Show that the commutator satisfies the Jacobi identity:

$$[A, [B, C]] + [B, [C, A]] + [C, [A, B]] = 0.$$

This is the operator analogue of one of the defining identities satisfied by the Poisson bracket.

This is the natural stopping point for this packet. The next step would be to ask how the classical bracket

$$\{f, g\}$$

is related to the quantum bracket

$$[\hat{f}, \hat{g}].$$

That question leads into canonical quantization and the correspondence

$$[\hat{f}, \hat{g}] \sim i\hbar \widehat{\{f, g\}}.$$

If you want this story to continue into Hamiltonian quantization, ask Dr. Linch and he can write the next installment.

10 Synthesis questions

i Synthesis 1: What generates time translations?

Explain, in a concise paragraph, why the Hamiltonian is the generator of time translations in classical Hamiltonian mechanics.

i Synthesis 2: Phase space versus configuration space

Explain the difference between configuration space and phase space. Why does the Hamiltonian naturally live on phase space?

i Synthesis 3: Classical and quantum brackets

Write a short paragraph comparing the Poisson bracket and the commutator. Your answer should include the phrases “antisymmetric,” “Jacobi identity,” “phase-space functions,” and “operators.”

11 Source note and further reading

This packet was adapted from a rough set of notes on time translations, Hamiltonian mechanics, and the Poisson bracket. The present version reorganizes the material into a guided derivation modeled on the minicourse packet *Renormalization Without Quantum Field Theory*.

Suggested reading links:

- [Hamiltonian mechanics](#)
- [Legendre transformation](#)
- [Phase space](#)
- [Poisson bracket](#)
- [Lie algebra](#)
- [Canonical quantization](#)